Early Human Development, 8 (1983) 351–375
Elsevier

EHD 00508

Infant incubators and radiant warmers

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Accepted for publication 15 May 1983

Summary

Incubators and radiant warmers are used to maintain the body temperature of newborn infants. This is best done so that the energy expended for metabolic heat production is minimized. The heat output of these devices is usually regulated by servocontrol to keep the skin temperature constant at a site on the abdomen where a thermistor probe is attached. In incubators, air temperature can also be controlled as an alternative to skin temperature servocontrol. Increased ambient humidity, heat shields and clothing have been used to decrease the evaporative or nonevaporative heat loss of infants in incubators under certain conditions. Double-walled incubators, by adding a second inner layer of Plexiglas, reduce radiant heat loss. They may also reduce total heat loss, but only if air temperature is controlled rather than skin temperature.

The minimal oxygen consumption under a radiant warmer is the same or perhaps slightly higher than it is for the same infant in an incubator. Compared with incubators, the partition of body heat loss is quite different under radiant warmers. Radiant warmers increase convective and evaporative heat loss and insensible water loss but eliminate radiant heat loss or change it to net gain. A heat shield of thin polyethylene film can be used with a radiant warmer to reduce heat loss by convection and evaporation. The major advantage of the radiant warmer is the easy access it provides to critically-ill infants without disturbing the thermal environment. Its major disadvantage is the increase in insensible water loss produced by the radiant warmer. Most infants can be safely and adequately cared for in either incubator or radiant warmer bed.

incubator; radiant warmer; heat loss; thermal environment; heat shield; newborn

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Keeping small babies warm has been shown to reduce their mortality [21,29,69] and morbidity [22,38,57] and to promote their growth [39]. Prior to the studies of Mordhorst [63], Day [28], and Brück [20], the premature newborn infant was thought to be a poikilotherm, whose body temperature responded passively to changes in environmental temperature. From these early studies [20,28,63], it was learned that the premature infant is not a poikilotherm, but rather a homeotherm, of limited capability to be sure, who can regulate his body temperature within a narrow range of ambient conditions.

As with all other homeothermic animals, the newborn infant produces heat within his body as a byproduct of metabolic processes and dissipates this heat to the surrounding environment by radiation, convection and conduction, and by evaporation of water from the skin and respiratory tract [42]. The metabolic heat production (M) equals the sum of heat losses to the environment by evaporation (E), radiation (R), convection (C) and conduction (K), plus the storage (S) of heat in the body:

$$M = E + R + C + K + S$$

This is the body heat balance equation [19], which is simply an expression of the First Law of Thermodynamics (conservation of energy). Under normal circumstances, when body temperature is not changing, heat storage is zero and endogenous heat production equals the sum of losses to the environment. Under certain conditions, the infant may gain heat from the environment by radiation (radiant warmer), convection (air warmer than skin), or conduction (heated mattress). In this case, the sum of the heat gained from the environment and the metabolic heat production equals the sum of heat losses by the other routes.

The newborn infant can prevent a fall of body temperature in either of two ways. He can increase heat production, primarily by nonshivering (chemical) thermogenesis, or he can decrease heat loss through vasoconstriction or postural changes. The ability to reduce heat loss is limited by the infant's large surface area in relation to the mass of heat-producing tissues, and the lack of thermal insulation [47,70]. Vasodilatation occurs in response to environmental warming, but the premature infant's defense against overheating is hampered by a lack of adequate sweat production [35,45].

Although nineteenth-century clinicians caring for premature infants recognized that cold stress increased their mortality [24,68], it was not until the clinical trials by Silverman and others [21,29,69] in the late 1950's and early 1960's that the survival of small infants was clearly shown to depend on the thermal environment. Premature or sick newborn infants are kept naked so that they may be closely observed. Moreover, nurseries are kept at temperatures below 25°C for the comfort of the adult staff. Thus, special devices must be used to create a more favorable thermal environment for such infants by providing an exogenous heat source and reducing heat loss to the environment. Two types of device are commonly used. The first to be used was the incubator, which provides an enclosed heated area around the infant. The heat is usually provided by forced circulation of air warmed by an electrical heater. The output of the heater is controlled manually by the operator or indirectly by thermostatic servocontrol to maintain either abdominal skin temperature or

incubator air temperature at a constant level. More recently, the radiant warmer has been introduced. The infant lies on an open bed surrounded by low walls and is warmed by an overhead panel that produces radiant heat. The heat output is regulated by servocontrol to maintain a constant abdominal skin temperature. In the following sections, incubators and radiant warmers will be discussed separately and then compared with regard to their advantages and disadvantages and their effects on the body heat balance of newborn infants.

Incubators

Thermal neutrality

The thermoneutral zone is defined as the range of ambient temperature within which metabolic rate is at a minimum, and within which temperature regulation can be achieved by nonevaporative physical processes alone [19]. The nonevaporative physical processes are vasomotor changes, piloerection and changes in posture, all means of changing the infant's thermal conductance [44]. Thus, for an infant in an incubator, the thermoneutral zone is defined in terms of incubator air temperature, but other environmental factors that influence heat loss must also be specified: (1) the pattern and velocity of air movement (constant for each incubator, but varies between incubator models), (2) humidity (depends on ambient humidity contributed by the incubator's humidification system, if used), and (3) the temperatures of the surfaces with which heat is exchanged by radiation, i.e. the inner walls of the incubator (depends on incubator air and room temperatures in a single-walled incubator, primarily on incubator air temperature in a double-walled incubator). The neutral thermal zone can also be defined in terms of the operative environmental temperature, which is a weighted average of mean radiant (or inner wall) temperature and air temperature [37]; or in terms of body skin temperature or the gradient between skin and environmental temperatures [72]. These alternate definitions still require that air velocity and humidity be specified.

The thermoneutral zone has been experimentally determined for newborn infants of varying body weight and postnatal age in several types of incubator under different conditions of humidity and radiant environment [15,43,46,67]. If ambient temperature is below the thermoneutral zone, the metabolic rate increases. The extra calories used to defend body temperature are not available for growth. If the ambient temperature exceeds the thermoneutral zone, evaporative water loss [15,66] or core temperature [76] increases. The smaller and less mature the infant, the narrower is the thermoneutral zone. The thermoneutral zone for very small premature infants in the first week of life may approach a single temperature point. With increasing postnatal age, the thermoneutral zone becomes cooler and wider [43,46].

Early use of incubators

Cone [24] has reviewed the earliest reported experiences with infant incubators, including the first published report of an incubator used for premature infants, by

Denuce in 1857 [30]. Silverman [68] has described the early experiences of Tarnier and Budin with infant incubators and Couney's use of the Tarnier incubator to care for premature infants exhibited at expositions and fairs in the early twentieth century.

In 1958 Silverman and coworkers [69] found that the survival rate of premature infants was greater at an incubator air temperature of 31 to 32°C than at 28 to 29°C. Even at 31 to 32°C, however, some of the smaller babies remained hypothermic [69]. This led to the development of an incubator in which the convective heating mechanism was supplemented by a radiant panel in the top that was controlled by a skin thermistor [1]. When this incubator was used to keep the abdominal skin at a temperature of 36°C, the survival of premature infants was even better than for those maintained in a convection incubator with air temperature 31 to 32°C [21,29].

Despite the promising results with this incubator, it was never manufactured on a large scale because of technical difficulties. Instead, new incubators were all of single-walled construction with fan-driven warm air convection and an optional servocontrol system for maintaining constant skin temperature. In these incubators a high air temperature is used to compensate for radiant heat loss to the relatively cool walls of the incubator. Double-walled incubators have been recently developed by several manufacturers, but single-walled incubators are still in widespread use.

Servocontrol

After the incubator of Agate and Silverman [1] was shown to increase survival when used to maintain a constant skin temperature of 36°C [21,29], most incubators were manufactured with the capability to maintain constant skin temperature by 'servocontrolling' heater output. Unlike the earlier incubator, however, the heat provided in response to skin cooling was provided by convection rather than radiation. The control point selected for abdominal skin temperature is usually between 36 and 37°C, based mainly on the work of Silverman et al. [71] with their combined convective-radiant incubator. Oxygen consumption at a skin temperature of 36°C was significantly lower than at 35°C, and lower (but not significantly) than at 37°C [71]. This evidence suggested that minimal metabolic rate (and thus thermoneutrality) occurred at a skin temperature between 36 and 37°C. This observation has led to the common clinical practice of maintaining skin temperature at 36.5°C, both with incubators and with radiant warmers.

In earlier incubators with servocontrol systems, the heater unit was either completely on, producing maximal heating, or off, depending on whether the thermistor probe temperature was below or above the selected control temperature. This resulted in sizable fluctuations in air temperature [10], which were found to result in increased apnea in premature infants [64]. All incubators currently being manufactured have proportional heating units, which have reduced the fluctuations in air temperature formerly seen with incubators whose heaters operated by on-off cycling.

The use of servocontrol to maintain a constant skin temperature seems to provide an easier means of approximating a thermoneutral environment in clinical practice, because the servocontrol system automatically adjusts the heat output to compensate

for changes in the rate of body heat loss. Skin temperature servocontrol does not allow monitoring of body temperature changes as signs of illness, but trends in air temperature can be used instead. For example, if the infant's thermoregulatory set point is increased by infection the air temperature required to maintain the same skin temperature will decrease. The use of skin temperature servocontrol requires careful nursing surveillance to check the site of attachment of the thermistor probe to the skin. If the probe is dislodged the infant can be overheated. If the probe becomes entrapped between the body and mattress or between two body surfaces, the heat output will diminish, and cooling will occur. The recommended site of attachment of the servocontrol probe is the skin in the midline of the upper abdomen in a supine infant. If the infant is in the prone position, the best site is not known, but the skin over the interscapular area should be avoided, as this area is rich in brown adipose tissue [2]. With increasing postnatal age, some infants become hyperthermic at a skin temperature of 36.5°C and require a lower servocontrol temperature.

The alternative method of regulating the heat output of older incubators is to manually adjust the heater control in response to changes in body temperature as measured periodically with a thermometer. At a given control setting, air temperature is maintained at a fairly constant level, although the controls do not indicate air temperature. Recently developed incubators provide a constant reading of air as well as skin temperature and offer servocontrol of air to a known temperature as an alternative to skin temperature servocontrol.

Humidity

Most incubators have a system for providing supplemental humidification, which can be used to raise the relative humidity to as high as 95% [11]. Increasing the humidity in the incubator reduces the heat loss by evaporation [74]. The effects of humidity on metabolic rate and total heat loss depend on study conditions. With air temperature held constant at a subthermoneutral level (31°C), Hey and Maurice [48] found that the total heat loss of term infants was 4% lower at moderate (45%) relative humidity and 11% lower at high (85%) humidity than it was at low (20%) humidity; metabolic rate was not significantly reduced at moderate or high humidity. At neutral air temperature (35°C), neither heat loss nor metabolic rate was affected by changes in humidity [48]. Sulyok et al. [74], however, found the metabolic rate and total heat loss of term infants to be lower at 50% relative humidity than at 20%, when air temperature was controlled to 32°C. At 80% humidity the metabolic rate was again higher than at 50%, but heat loss was not significantly different [74]. If the skin is servocontrolled to a constant temperature, the decrease in evaporative heat loss with higher humidity might be balanced by an increase in nonevaporative heat loss as a lower air temperature is needed to maintain the same skin temperature. Belgaumkar and Scott [14] found that premature infants who were servocontrolled to keep the abdominal skin at 36°C had lower rectal temperatures in low humidity than at high humidity. Metabolic rate was not measured. One interpretation of these results would be that the infants were

subthermoneutral and that their metabolic response was inadequate to maintain core temperature in the dry environment. When evaporative heat loss was decreased by added humidity, the metabolic response was apparently adequate to defend core temperature. This difference would probably not be observed if a higher skin temperature were selected, providing a thermoneutral environment. An alternative explanation of the results of Belgaumkar and Scott [14] is that the thermistor probe might have impeded evaporation from the skin directly beneath the probe, producing false high skin temperature at low humidity, and consequently lower rectal temperature.

In summary, increased relative humidity reduces evaporative heat loss under all circumstances. The impact of this effect on overall thermal balance depends on other conditions. There seems to be little or no effect of humidity on metabolic rate or total heat loss under thermoneutral conditions. However, when stressed by subthermoneutral conditions, infants will benefit from increased humidity by reducing total heat loss. The other effect of increased humidity is to reduce insensible water loss [41,45,74].

Although the effects of humidity on heat and water balance are well recognized, many clinicians avoid using supplemental humidity because of concern that pseudomonas and other similar pathogenic bacteria may grow in the water reservoir.

Heat shields

Plexiglas heat shields reduce the radiant heat loss of infants in single-walled incubators by providing a warmer surface for radiant heat exchange [18,49,50]. A Plexiglas heat shield with closed ends reduces the insensible water loss of premature infants with birth weight below 1.25 kg who are nursed in single-walled incubators [33]. Such heat shields have been shown to reduce the oxygen consumption of term infants in single-walled incubators with air temperature controlled to 32°C, which was 10 to 12°C above room temperature [50]. However, when incubators were servocontrolled to maintain abdominal skin temperature at 36.5°C the oxygen consumption of premature infants was not reduced by a Plexiglas heat shield [18].

A blanket made of double-layer plastic packing with 'bubble' air pockets between the layers has been found to reduce the insensible water loss [58] but not the oxygen consumption [27] of infants in single-walled incubators that are servocontrolled to maintain a constant skin temperature.

Both Plexiglas heat shields and thermal blankets seem to reduce insensible water loss (and evaporative heat loss) of premature infants in single-walled unhumidified incubators. They also tend to reduce radiant and convective heat losses. However, the effect on total heat loss seems to be minimal in incubators operated by skin temperature servocontrol. Apparently, the heat shield or blanket reduces the amount of exogenous heat required to maintain the same skin temperature, so the incubator heat output is reduced. Under these circumstances it would appear that the incubator expends fewer calories (and consumes less electrical energy), but the energy expenditure of the infant is not reduced by the shield or blanket. The slight reduction in evaporative heat loss is presumably offset by an increase in nonevaporative losses as the incubator cools.

Less is understood about the effects of clothing on the heat and water balance of infants in incubators. Levine et al. [55] compared the effects of clothing of different materials on the measurement of insensible water loss of infants. Hey and coworkers [47,51] studied the effects of clothing on the heat balance of newborn infants in a metabolic chamber under conditions simulating a conventional incubator. They calculated the increase in specific thermal insulation that resulted from the clothing and showed that nonevaporative heat loss was reduced by clothing when environmental temperature was constant [47]. Using the same metabolic chamber, Stothers [73] found that the oxygen consumption and total heat loss of otherwise naked term infants was reduced when wearing a hat, but not when a similar-sized area of the abdomen was covered with the same material. There is no information regarding the effects of clothing on heat and water losses of premature infants in incubators servocontrolled to maintain constant skin temperature.

Double-walled incubators

In an attempt to reduce radiant heat loss by infants in incubators, several manufacturers have introduced incubators with walls made of two layers of Plexiglas separated by a layer of air several centimeters thick. The inner surface of the inner wall should be warmer and less responsive to changes in ambient nursery temperatures, thus reducing the infant's heat loss by radiation. Several studies have examined the effects of a double-walled incubator on the body heat loss of premature infants [16,60,82].

Clinical studies of double-walled incubators

Yeh et al. [82] measured respiratory gas exchange and insensible water loss of ten premature infants in single and double-walled (Ohio Medical Products, Intensive Care Incubator) incubators. Oxygen consumption, carbon dioxide production and insensible water loss were all significantly reduced in the double-walled incubator. The mean abdominal temperatures were below 36.0°C in both incubators. Moreover, the abdominal skin temperatures were significantly lower in the single-walled incubator, suggesting that the infants in the single-walled incubator were subthermoneutral and thus had higher rates of oxygen consumption and carbon dioxide production than in the double-walled incubator. The reduction in insensible water loss observed by Yeh et al. [82] in the double-walled incubator is difficult to explain, unless it was due to major differences in incubator humidity or air velocity, which were not measured.

Marks et al. [60] measured the oxygen consumption of ten premature infants in a double-walled incubator (Air-Shields C-86 Double Wall Isolette) and in the same incubator with the inner wall removed. They reported a reduction in oxygen consumption in the double-walled incubator. The heat output was manually adjusted to maintain incubator air within the thermoneutral zone as taken from published tables. Air temperatures were the same in both incubators. One would not expect the same air temperature to be neutral for a given infant in both single and double-walled incubators. The neutral air temperature in the single-walled incubator should be

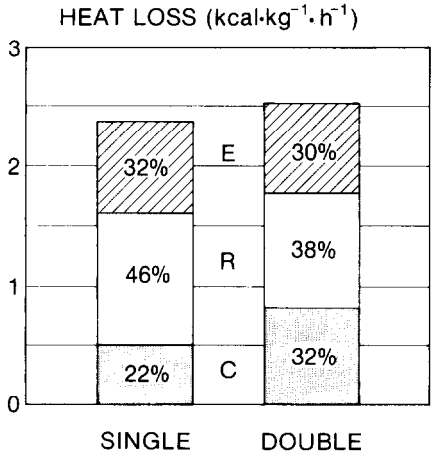


Fig. 1. Partition of body heat loss of premature infants in single (SINGLE) and double-walled (DOUBLE) incubators. E = evaporation, R = radiation, C = convection (+conduction). Data from Bell and Rios [16].

higher because of the cooler walls. The subjects of Marks et al. were in cooler operative environments in the single-walled incubator and were possibly below the thermoneutral zone, thus explaining their higher rate of oxygen consumption.

We have conducted a similar study [16] in eight premature infants using the Air-Shields Double Wall Isolette (model C-86) with and without the inner wall, but the incubator in both cases was regulated by servocontrol to keep the abdominal skin at 36.5°C. Using skin temperature servocontrol in this fashion, the mean operative environmental temperature was the same (33.0°C) in both incubators. There were no differences in body temperatures. The mean incubator wall temperature was higher in the double-walled incubator, but the incubator air temperature was lower (Table I). Oxygen consumption, carbon dioxide production and insensible water loss were not significantly different. The total rate of body heat loss was similar, 2.37 kcal (9.92 kJ)·kg^{-1}·h^{-1} in the single-walled incubator and 2.52 kcal (10.55 kJ)·kg^{-1}·h^{-1} with the double wall. The only difference was in the partition of body heat loss. The evaporative heat loss was unchanged; but radiant heat loss was reduced and convective loss increased in the double-walled incubator (Table I). The changes in the partition of heat loss are shown in Figure 1.

Our findings suggest that a double-walled incubator offers no advantage over a servocontrolled single-walled incubator in the overall thermal economy of premature infants. When called upon to maintain the same abdominal skin temperature, the single and double-walled incubators produced the same operative environmental temperature (33.0°C). Adding the second, inner wall raised the inner wall temperature but lowered the incubator air temperature, so that the same thermal gradient from skin to environment was maintained. When servocontrolled in this fashion, the incubators allowed the infants to remain in thermal equilibrium at the same plane of

TABLE I

Heat exchange data for premature infants in single and double-walled incubators [a]

	Single	Double	P-value [b]
Temperatures (°C)			
Abdomen	36.4 (0.2) [c]	36.3 (0.3)	ns
Mean skin	36.3 (0.3)	36.2 (0.3)	ns
Rectum	37.2 (0.1)	37.2 (0.1)	ns
Incubator air	34.8 (2.1)	33.8 (1.7)	0.008
Mean incubator wall	31.7 (1.1)	32.3 (1.2)	0.021
Operative environmental	33.0 (1.4)	33.0 (1.4)	ns
Gas exchange ($ml \cdot kg^{-1} \cdot min^{-1}$)			
Oxygen consumption	7.95 (1.16)	8.40 (1.77)	ns
Carbon dioxide production	7.33 (1.43)	7.64 (1.76)	ns
Insensible water loss ($g \cdot kg^{-1} \cdot h^{-1}$)	1.32 (0.43)	1.29 (0.33)	ns
Heat exchange ($kcal \cdot kg^{-1} \cdot h^{-1}$) [e]			
Metabolic heat production	2.36 (0.37)	2.49 (0.52)	ns
Total heat loss	2.37 (0.41)	2.52 (0.55)	ns
Evaporative heat loss	0.76 (0.25)	0.75 (0.19)	ns
Radiant heat loss	1.10 (0.17)	0.95 (0.20)	0.006
Convective heat loss [d]	0.51 (0.44)	0.82 (0.40)	0.001

[a] From Bell and Rios [16].
[b] Two-tailed t-test for paired observations. ns indicates $P \geq 0.05$.
[c] Mean (standard deviation).
[d] Includes conductive heat loss.
[e] Conversion to SI units: 1 kcal = 4.187 kJ.

heat production, since the total rate of body heat loss was the same.

If skin temperature servocontrol is not used, then a double-walled incubator reduces body heat loss compared to a single-walled incubator at the same air temperature [60], because radiant heat loss is reduced while evaporation and convection are unchanged. This difference could be readily overcome, however, by raising the air temperature in the single-walled incubator.

Other studies of double-walled incubators

When a stable thermal environment is maintained by keeping the door closed, the double-walled incubator offers no advantage when operated by skin temperature servocontrol. However, it does allow a smaller fall in operative environmental temperature when the door is opened, since the warm inner wall acts as a heat source and cools more slowly than the air temperature. Figure 2 shows the change that occurred in operative environmental temperature when the door was opened for 20 min on three incubators that were set to maintain a constant air temperature of 36°C. The operative temperature in the Air-Shields C-86 single-walled Isolette fell by about 5°C. The Air-Shields C-100 Isolette is a newer model that employs an 'air curtain' to redirect the flow of warm air upward across the front of the incubator

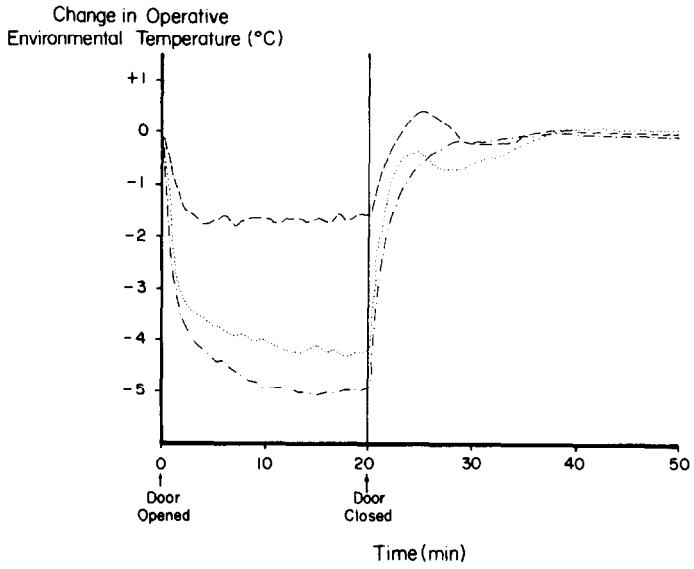


Fig. 2. Effect of opening and closing the incubator door (access panel) on operative environmental temperature. The dashed line (— — —) is the temperature in the Air-Shields C-100 double-walled Isolette. The dotted line (· · · · · ·) represents the Air-Shields C-100 single-walled Isolette, and the alternating dashed-dotted line (· – · – ·) is the Air-Shields C-86 single-walled Isolette.

when the door is opened. As can be seen from Figure 2, the operative temperature fell by about 4°C in the C-100 single-walled Isolette, and by less than 2°C in the C-100 double-walled Isolette.

The Ohio IC Incubator (Ohio Medical Products) offers several design features that are different from the Air-Shields double-walled incubators (Isolette C-86 and C-100). The Air-Shields incubator hood is shaped as a pentagonal cylinder. The heated air flows from the left end of the hood floor up over the infant, within the inner wall, and reenters the heating compartment at the right end of the floor. The Ohio IC is a double-walled incubator with the hood shaped as a circular cylinder. The heated air passes between the walls from back to front, and about 10% of the heated air is diverted through the inner wall into the infant chamber. When the front of the hood is opened by retracting part of the cylinder into itself, the entire stream of heated air is directed into the inside toward the infant. Both the Ohio IC and the Air-Shields C-100 offer two modes of regulating heater output. Both can be operated by proportional servocontrol to maintain constant abdominal skin temperature. In addition, the C-100 can be servocontrolled to maintain constant air temperature, either with the standard air temperature probe below the infant compartment or with an auxiliary air probe suspended 16 cm above the center of the mattress. This air temperature servocontrol mode has replaced the manual control mode (a less sophisticated form of air temperature servocontrol) that was used on the old C-86

TABLE II

Performance characteristics of three double-walled incubators

	Air-Shields C-86		Air-Shields C-100		Ohio IC
Warm-up time (to 35°C), min	50 (7) [a]		38 (2)		26 (2)
Overshoot, °C	0.3 (0.2)		0.0 (0.0)		0.8 (0.1)
Time to equilibrium [b], min	58 (10)		44 (7)		86 (30)
Stability (90–150 min), range, °C					
Air temperature	0.2 (0.1)		0.2 (0.1)		0.4 (0.2)
Mean wall temperature [c]	0.1 (0.1)		0.1 (0.1)		0.5 (0.1)
Effect of opening door	Ports only	Access panel	Ports only	Access panel	First stop (14 cm)
Air temperature drop, °C					
1 min	0.7	3.6	0.3	1.8	0.3
5 min	0.3	5.1	0.3	2.7	0.8
10 min	−0.1	5.1	0.0	2.6	0.3
20 min	0.1	5.0	0.1	2.4	−0.8
30 min	0.1	5.0	0.1	2.1	−1.0
Electrical power consumption, W					
Maximum	218		275		465
Minimum	10		17		34
Air velocity, m/s					
Door closed	< 0.05		< 0.05		≤ 0.06
Door open	< 0.05		< 0.05		≤ 0.09
Fresh air intake, l/min	35		26		133
Sound level, dB(A)	55.0		56.6		55.0

[a] Mean (standard deviation).
[b] Range of air temperature fluctuation less than 0.5°C.
[c] Mean of front and rear inner wall temperatures.

incubator. The alternative operating mode of the Ohio IC incubator, designated as the 'manual' mode, is designed to maintain a constant 'environmental' temperature; the 'environmental' temperature is the average of the front and rear inner wall temperatures. We have tested several performance characteristics of three double-walled incubators, the Air-Shields C-86 Double Wall Isolette, the Air-Shields C-100 Double Wall Isolette (with the 'air curtain'), and the Ohio IC Incubator [17]. All studies were performed in triplicate using a single incubator of each model. Studies were performed in the same fashion under identical circumstances with all three models and, whenever possible, were performed simultaneously. All studies were performed with room temperature between 21 and 25°C. The results are summarized in Table II.

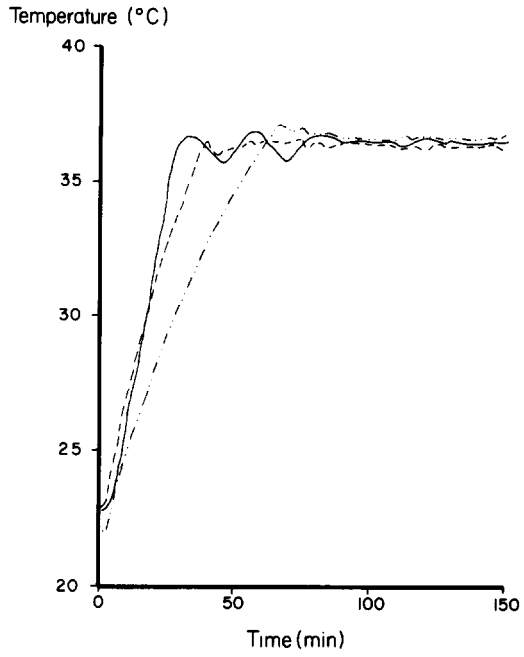


Fig. 3. Warm-up and stability test of three double-walled incubators. Incubators were rapidly warmed from room temperature to control temperature 36.0°C. The Air-Shields C-100 double-walled Isolette (dashed line, — — —) and the Air-Shields C-86 double-walled Isolette (alternating dashed-double dotted line, – · · – · ·) were both regulated by air temperature servocontrol, with the sensor probe suspended 16 cm above the center of the mattress. The Ohio IC Incubator (solid line, ————) was regulated in the 'manual' control mode.

In the first experiment, the incubators were turned on at room temperature and immediately set to maintain air temperature at 36°C. In the C-86 this was done by suspending the servocontrol probe 16 cm above the center of the mattress. The C-100 was operated in the air temperature control mode with the auxiliary probe also 16 cm above the mattress. The Ohio IC was operated in the 'manual' mode. The results of one of the warm-up trials are shown in Figure 3. The Ohio IC had the shortest warm-up time but required the longest time to reach an equilibrium temperature. The Ohio incubator also showed the largest fluctuation in air temperature once equilibrium was reached. This may be important in infants who are prone to apnea, since increased frequency of apnea has been shown to be associated with fluctuations in incubator air temperature [64].

In the second experiment the effects of opening the incubator doors or ports were studied. The Air-Shields incubators were tested both with the two front ports open and with the entire access panel open. The Ohio incubator was tested with the hood open to the first stop (14 cm). The results are shown in Table II and in Figure 4. With the ports open the air temperature in the Air-Shields incubators dropped by less than 1°C and returned to baseline within 10 min. The mean wall temperature

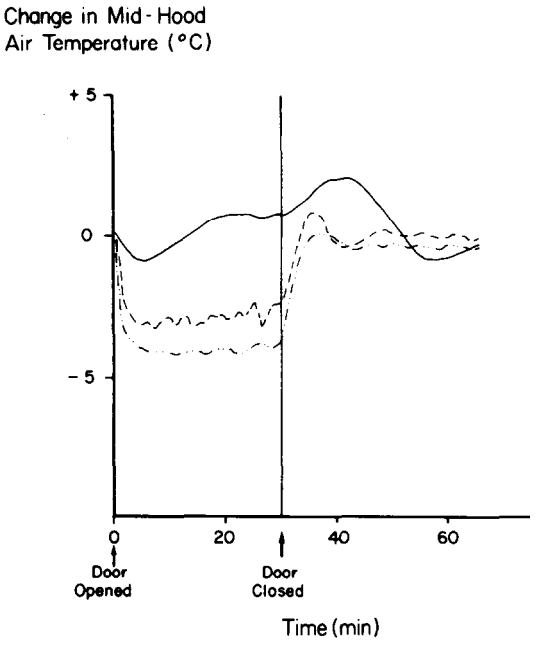


Fig. 4. Effect of opening and closing the incubator door on mid-hood air temperature. The dashed line (— — —) is the air temperature in the Air-Shields C-100 double-walled Isolette, and the alternating dashed-double dotted line (– · · – · ·) is the Air-Shields C-86 double-walled Isolette, both with the front access panel opened. The solid line (————) is the air temperature in the Ohio IC Incubator with the hood opened to the first stop (14 cm).

changed even less. Figure 4 shows the results of a typical experiment with the access panels open on the Air-Shields incubators. With the access panel open the air temperature dropped by about 4°C in the C-86 and by about 3°C in the C-100 (Fig. 4). The air temperature returned to baseline within about 10 min after the panel was closed in both Air-Shields incubators. The air temperature in the Ohio incubator fell less than 1°C, then increased to about 1°C above baseline (Fig. 4). When the door was closed the air temperature rose another 1°C before returning to baseline in about 20 min. These results must be carefully applied to clinical care situations. Some infant care tasks that can be performed in the Ohio incubator with the hood at the first stop can be done through the Air-Shields ports, while others require opening the access panel on the Air-Shields incubators.

The maximum and minimum electrical power consumed were measured with a watt-meter. The minimum power is the amount required to operate the fan and the indicator lights. The minimum was 10 W for the C-86, 17 W for the C-100, and 34 W for the Ohio IC. The maximum power levels recorded for these units were 218 W, 275 W, and 465 W for the C-86, C-100, and Ohio IC, respectively.

Air velocity was measured 6 cm above the center of the mattress using a hot-wire anemometer, with the probe parallel to the long axis of the incubator hood. The air

velocity was low in all incubators. It was below the sensitivity limit of the anemometer, 0.05 m/s, in both Air-Shields incubators, even with the access panel open. The maximum air velocity detected in the Ohio incubator was 0.06 m/s with the hood closed and 0.09 m/s with the hood open to the first stop (14 cm). Fresh air intake was measured by oxygen dilution [9]. It was 35 l/min in the C-86, 26 l/min in the C-100, and 133 l/min in the Ohio IC. Sound levels were similar in the three incubators (Table II).

In the usual nursery environment, the double-walled incubator seems to reduce the fall in operative environmental temperature that occurs when the door is opened, but offers little or no advantage over a servocontrolled single-walled incubator when the thermal environment is not disturbed. However, the thermal insulation provided by the air between the double Plexiglas walls should offer extra protection against the rapid fluctuations in ambient air temperature and the cold outside air that are encountered during transport of small infants between hospitals. In such a situation, the inner wall surface of a single-walled incubator would cool faster, and the ability of the heater to compensate for increased radiant heat loss by raising air temperature would be exceeded sooner than with a double-walled incubator.

Safety considerations

In the United States, all medical devices are regulated by the Bureau of Medical Devices, a branch of the Food and Drug Administration [53]. In addition, the Association for the Advancement of Medical Instrumentation has drafted safety and performance standards for infant incubators [7]. A number of new safety features have been incorporated in incubators manufactured in recent years [9]. Most incubators are now equipped with a separate safety thermostat that prevents overheating. There are also alarms that indicate over or under heating, loose probes, and power failure. The heater controls are less susceptible to accidental adjustment. Some older models had heater dials that did not lock and could easily be rotated by incidental contact. The Air-Shields C-100 has thumbwheel controls that are not easily changed by accident, and the Ohio IC has touch-sensitive controls that require several seconds of sustained contact. Despite these safety features rare episodes of dangerous malfunction still occur [40]. It is essential that routine bioengineering maintenance for infant incubators include testing of all safety functions.

Another potential hazard that has been linked to infant incubators is exposure to mercury vapor released from broken thermometers or mercury expansion switches [61,75]. Since mercury thermometers are routinely used to monitor body temperature in most nurseries and are sometimes broken, it is difficult to know how much of the mercury detected in incubators might be from that source rather than the mercury thermometers and switches in the incubators. Whatever the source, mercury vapor contamination is a potentially serious problem, the scope of which is unknown.

Radiant warmers

Early use of radiant warmers

The incubator designed by Agate and Silverman [1] was the first attempt to use an electrical source of radiant energy to warm infants. One of the first commercial heating devices to employ a radiant heat source was the Sierracin cradle warmer [4], an open-end incubator with a clear plastic hemicylindrical canopy containing a radiant heat source. Several clinical studies [36,56,80] reported results obtained with the Sierracin warmer. Levison and coworkers [56] compared body temperatures and oxygen consumption of infants in the Sierracin warmer, servocontrolled to keep the abdominal skin at 36.7°C, with the same measurements in a convectively-heated incubator servocontrolled to 36.5°C. Body temperature was maintained in both units, but oxygen consumption was slightly higher in the convectively-heated incubator, presumably because of the lower abdominal skin control temperature. Other studies [36,80] showed that the Sierracin warmer could be safely used to keep infants warm if servocontrolled to maintain constant abdominal skin temperature. Friedman et al. [36] compared abdominal control temperatures of 36.2, 36.5, and 37.1°C. At 36.2°C infants became restless and some became hypothermic. Oxygen consumption and core-skin temperature gradient were lowest at 37.1°C, and the infants were least active and seemed most comfortable at 37.1°C. In contrast, Yashiro et al. [80] concluded that 36.7°C was the best skin temperature, because some infants became hyperthermic at 37.2°C. The optimal skin temperature for control of radiant heaters is still not known. Nor do we know how much this optimal temperature varies among infants. The Sierracin warmer became outmoded as other radiant warmers were designed that offered easier access to the infant and improved safety features [4].

Radiant warmers with overhead heating panels were first used to warm infants during resuscitation in the delivery room [32]. One of the first units of this type, the Merco Infant Warmer [32], had no safety alarms and lacked stability against tip-over [4]. It has subsequently been replaced by other, safer units [4,5].

Heat sources, control properties, and safety features

Radiant heaters now in use provide heat in the form of infrared energy that is almost completely absorbed by the skin to a depth of less than a millimeter [4]. The heat source varies, but may consist of infrared lamps or a resistance wire in a quartz tube. Most radiant warmers regulate heat output by proportional servocontrol to maintain the abdominal skin at the chosen control temperature. A non-servocontrol mode is offered as an alternative on some units. Since there is no feedback of information about body temperature, this mode increases the risks of hypothermia and hyperthermia. Radiant warmers should be used only in the skin temperature servocontrol mode.

Several detailed evaluations of the performance and safety of radiant warmers have been published [4–6]. The Association for the Advancement of Medical

TABLE III

Design criteria for radiant warmers

Heating element

1. Produces sufficient uniform heat to warm small infant
2. Far infrared (wavelength greater than 3 μm), with irradiance at infant's eyes less than 0.3 W/cm^2
3. Minimal radiation in near infrared at wavelengths less than 12 μm
4. Negligible radiation of visible light
5. Heating element protected from accidental contact by patient or personnel
6. Comfortable for personnel to work under

Control unit

1. High and low temperature alarms
2. Defective probe alarms
3. Minimizes temperature excursions
4. Appropriate warning placards
5. Infant's temperature indication
6. Control point setting visible
7. Power indication
8. Low leakage current
9. Radio frequency compatibility
10. Overcurrent protection
11. Insensitive to line voltage variation
12. Self-check provisions

General mechanical characteristics

1. Tipover resistance
2. Secure bassinet side panels
3. No sharp edges
4. Rugged construction
5. Good mobility
6. Easy to clean

Bassinet and Accessories

1. Convenient work space and access to infant
2. Availability of resuscitator, aspirator, oxygen
3. Adequate lighting
4. Storage space
5. Easy use of x-ray, phototherapy lamps
6. Mattress tilting provisions

From ref. 4.

Instrumentation has also published recommended performance and safety standards for radiant warmers [8]. Table III lists criteria that should be considered in the design or evaluation of radiant warmers for infants. The high and low temperature alarms and the defective probe alarm are particularly important. One of the principal dangers of radiant warmers is infant hyperthermia due to a loose servo-control probe. Most new models incorporate an alarm that is activated when a dislodged probe produces a rapid fall in probe temperature. Overheating is still possible if the alarm can be silenced without prompt investigation of the cause of the alarm. Because of the potential risk of hyperthermia, infants under radiant warmers

require more frequent monitoring of axillary or rectal temperature and closer nursing surveillance than infants in incubators.

The danger of hyperthermia and other reported and theoretical risks of radiant warmers have been reviewed by the Committee on Environmental Hazards of the American Academy of Pediatrics [3]. The Committee recommended that special attention be given to the adequacy of safety mechanisms when choosing a radiant warmer to be purchased [3]. They also emphasized the need for close monitoring of body temperature and fluid balance of infants under radiant warmers. Another risk that has been reported is first-degree burns caused by heating of plastic-linked disposable diapers in contact with the skin [34].

In spite of these risks, radiant warmers have gained widespread acceptance. They provide open access to infants needing resuscitation or procedures while still providing sufficient exogenous heat to replace body heat losses. Radiant warmers are routinely used in most delivery rooms for infant resuscitation. They are also used in many nurseries to care for small premature infants or critically-ill term infants who require closer monitoring and more frequent procedures.

A new heating device for infants has recently been designed to incorporate features of both enclosed incubator and radiant warmer [6]. The Puritan-Bennett Infra-Care 2000 has been carefully tested; despite enthusiasm about the 'superior and innovative' design concept, the unit was not recommended because of a number of problems not encountered with the better conventional incubators and radiant warmers [6].

Modifications to reduce convection and evaporation

Radiant warmers reduce or eliminate heat loss from the infant's body by radiation but increase convective and evaporative losses. Several types of heat shield or thermal blanket have been used in attempts to prevent the increased heat losses by convection and evaporation. Yeh and coworkers [81] showed that the insensible water loss of infants under radiant warmers could be reduced by placing a rigid Plexiglas heat shield over the infant. Other investigators [13,18] have failed to confirm this finding and have advised against the use of Plexiglas heat shields with radiant warmers. The Plexiglas absorbs the infrared energy radiated by the warmer and prevents it from directly reaching the infant [13,18]. Some of the heat absorbed by the Plexiglas may be subsequently radiated to the infant's body, but this would presumably be less effective than direct radiation from the heating element [13]; the Plexiglas heat shield might also interfere with the warmer's servocontrol mechanism, since changes in heater output in response to skin temperature changes would not be immediately sensed at the skin level. In one study [18], the abdominal temperature was lower with the heat shield in spite of increased heater output, and some infants could not achieve the abdominal control temperature of 36.5°C.

Baumgart and coworkers [12,13] have tested several types of convective heat shield. A blanket made of thin plastic material used for wrapping food was found to be the most effective. It did not block radiant heat uptake by the infants but reduced their losses by convection and evaporation [12,13]. The authors apparently noted no

skin burns of the type reported by Fleischman [34]. A technique used in some nurseries is to stretch the thin plastic food wrapping material across the top of the bassinet portion of the warmer, from wall to wall. This technique should reduce convective heat loss and might also reduce evaporation somewhat by decreasing air movement over the skin surface. Thin polyethylene (Glad Wrap, Union Carbide) is the preferred material; it transmits more of the infrared energy radiated from the warmer than does Saran Wrap [54]. The use of this thin polyethylene film with radiant warmers has proved to be especially valuable in the thermal care of very small premature infants.

Incubator versus radiant warmer

Accessibility

A major advantage of the radiant warmer is the open access it provides to infants who require procedures, such as endotracheal intubation, blood sampling, and insertion of catheters or needles for intravenous infusions. The radiant warmer allows these procedures to be performed with no change in skin temperature, as long as the sensor probe is correctly attached and the radiant heat is not blocked by drapes or the head and hands of physician or nurse. The air and skin temperatures in an incubator are well maintained if procedures can be performed through the two circular ports that are present on the front of most incubators. If, however, the entire front panel of an Air-Shields incubator must be opened, the air temperature falls by 3 to 5°C (Table II); air temperature returns to the previous level within 10 min after the panel is closed (Fig. 2). The abdominal skin temperature of a one-week-old infant weighing 0.8 kg fell by 0.3°C when the access panel of an Air-Shields C-86 Double Wall Isolette was opened (Fig. 5). The Ohio IC double-walled incubator, in comparison, maintained skin temperature by increasing air temperature when the door was opened to the first stop (14 cm) (Fig. 5). When the door was opened to the second stop (90°), as would be necessary for endotracheal intubation, for example, air temperature fell by 3°C below the starting point and skin temperature fell by 1.6°C (Fig. 5). Both incubators were servocontrolled to keep the abdominal skin at 36.3°C. These changes in skin temperature can be avoided with radiant warmers, even during procedures.

It was thought that the exposure of infants under radiant warmers to unfiltered room air might lead to increased bacterial colonization. Two studies [23,62], however, have demonstrated no such increase.

Rewarming after hypothermia

Robinson and Jones [65] have shown that infants admitted to the nursery with core temperatures below 35°C could be warmed to normal body temperature more rapidly with radiant warmers than in incubators.

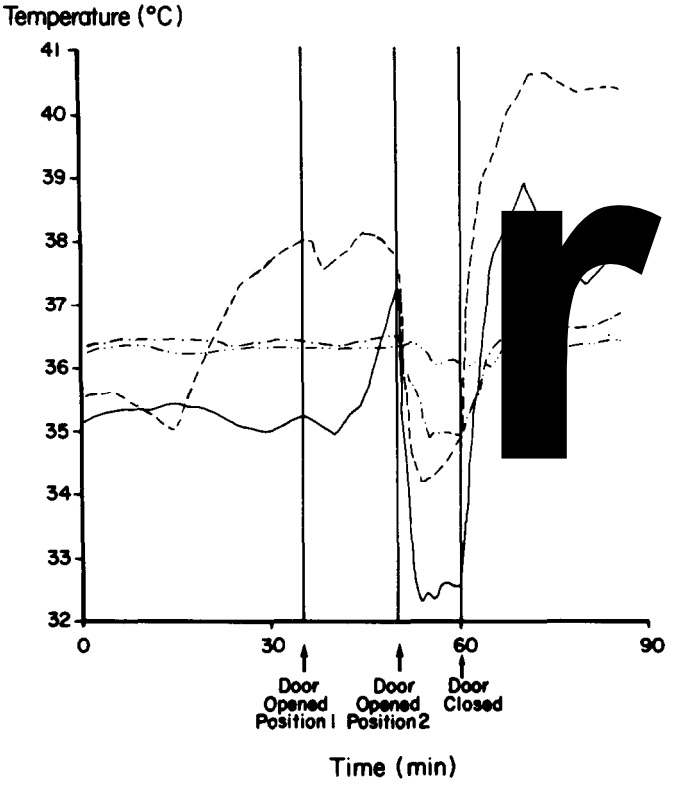


Fig. 5. Effect of opening and closing the incubator door on incubator air temperature and abdominal skin temperature of a 1-week-old 0.8 kg infant. The dashed line (— — —) and the alternating dashed-double dotted line (–··–··) represent the air and skin temperatures, respectively, in the Air-Shields C-86 double-walled Isolette. The solid line (————) and the alternating dashed-dotted line (–·–·) represent the air and skin temperatures, respectively, in the Ohio IC Incubator. Both incubators were regulated by servocontrol to keep the abdominal skin at 36.3°C. Position 1 is with the two front ports of the Air-Shields incubator opened and with the hood of the Ohio incubator opened to the first stop (14 cm). Position 2 is with the entire front access panel of the Air-Shields incubator opened and with the hood of the Ohio incubator opened to the second stop (90°).

Insensible water loss

The rate of insensible water loss of newborn infants has been reported to be higher under radiant warmers than in incubators, by 40 to 190% depending on the type of warmer and the size of the infant [18,78,79]. Several of these studies [18,79] were done using the Potter scale, which can register erroneously high weight loss if the scale temperature is increasing [25,31]. Thus, falsely high estimates of insensible water loss would be expected if the scale were not adequately prewarmed before each study. The study [79] reporting the largest increase in insensible water loss with the radiant warmer was performed with a Potter scale, but sizable increases have

also been reported using water balance calculations [78] and other indirect methods of estimating insensible water loss [52]. Although the magnitude of the increase in insensible water loss under the radiant warmer may have been overestimated in some previous studies, there is no doubt that the increase occurs. Most data indicate that the increase in insensible water loss with the radiant warmer (without phototherapy) is between 40 and 100%, not up to 190% as previously suggested [79].

With radiant warmers, the insensible water loss cannot be easily reduced by raising the humidity, as it can in incubators. The increased insensible water loss produced by the radiant warmer must be anticipated and the water intake increased accordingly. When a large premature or term infant is moved from an unhumidified incubator to a radiant warmer, the water intake should be increased by about 20% (40% of losses insensible × 50% increase) while body weight change, urine volume and osmolality (or specific gravity), and serum sodium concentration are carefully monitored. In small premature infants a larger fraction of water loss is insensible, so the initial increase in water intake with a radiant warmer should be 40 to 50%. The decrease in total water requirement when a small premature infant is moved from radiant warmer to incubator would be the same absolute amount but a smaller percentage, approximately 30% (40/140 or 50/150). These suggested changes in water intake are rough guidelines that will be subsequently adjusted according to body weight change, urine volume and osmolality, and serum sodium concentration. Although the increased water loss is a disadvantage for the radiant warmer, the risk of dehydration can be minimized by careful monitoring of body water balance.

Minimal oxygen consumption

Levison et al. [56] were the first to compare the oxygen consumption of infants under radiant heaters with those in convectively-heated incubators. In their study the oxygen consumption was slightly but significantly higher in the incubator. This difference can probably be explained by the difference in abdominal skin control temperature, 36.5°C in the incubator and 36.7°C under the radiant heater [56].

Subsequently, three studies [18,26,59] showed that the resting oxygen consumption of infants under radiant warmers was not significantly different from the resting oxygen consumption of the same infants in incubators, when both types of heating device were servocontrolled to maintain abdominal skin at the same temperature (36.5°C).

Noting that the mean resting oxygen consumption was higher (by 1.6 to 4.7%), though not significantly so, in each of these three studies [18,26,59], LeBlanc [54] reasoned that a larger sample of infants might demonstrate a statistically higher oxygen consumption under the radiant warmer. With both heating systems servocontrolled to keep abdominal skin temperature at 36.0°C (lower than previous studies), he found the oxygen consumption of 16 premature infants to be 8.9% higher under the radiant warmer, a difference that was statistically significant [54]. Although statistically significant, the clinical importance of this observation can be questioned, since the reduction in oxygen consumption of 0.6 ml/kg per min in the incubator corresponds to a savings in 24-h energy expenditure of only 4 kcal (17 kJ)

per kg. All of these studies have ignored the possibility that the skin temperature associated with the minimal oxygen consumption, i.e. the 'neutral' skin temperature, might be different under a radiant warmer than in an incubator. In summary, there appears to be little if any extra energy expended by infants under radiant warmers. This suggests that their total heat loss, i.e. the sum of net heat exchanges with the environment by evaporation, radiation, convection and conduction, is not increased by the radiant warmer [77].

Partition of body heat loss

Although the total body heat loss is not increased by the radiant warmer, the partition of body heat loss into evaporation, radiation, convection and conduction is very different than it is in the incubator [18,77]. The modest increase in evaporative heat loss under the radiant warmer is accompanied by a larger increase in heat loss by convection and a change from net loss to net gain of heat by radiation [77] (Fig. 6). If the increase in water loss by evaporation is countered by increased water intake, the difference in the partition of body heat loss is probably unimportant, because the demand for metabolic heat production is not increased by the radiant warmer. Apart from the effects on water balance of changes in evaporation, it seems that the newborn infant is metabolically indifferent to changes in the partitioning of body heat loss among the four routes, as long as the sum of net exchanges of thermal energy by these routes between the infant's body and the environment is unchanged [16].

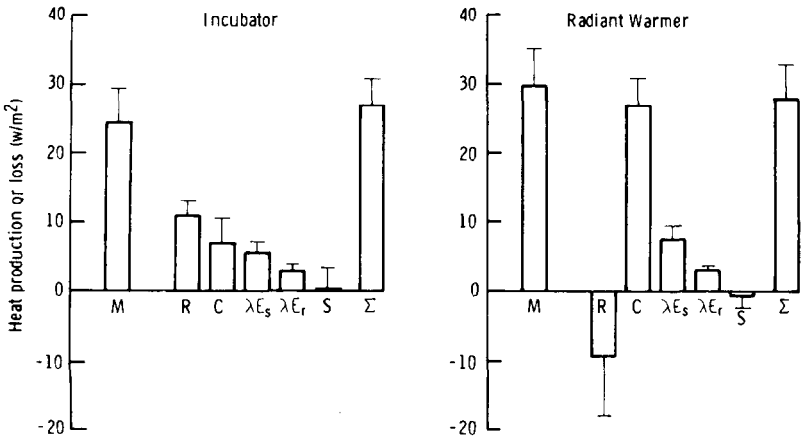


Fig. 6. Mean (± S.D.) values of metabolic heat production (M); heat loss by radiation (R), convection (C), skin evaporation (λE_s) and respiratory evaporation (λE_r); heat storage (S); and total heat loss ($\Sigma = R + C + \lambda E_s + \lambda E_r$). Negative values indicate heat gain. From Wheldon and Rutter [77].

Summary

The major advantage of radiant warmers over incubators is the easy access they provide to critically-ill infants without disturbing the thermal environment. This is especially important in the delivery room and during the first few days of life. Moreover, some very small premature infants cannot be maintained in incubators, even at maximal air temperatures, because of their high rate of heat loss by evaporation [77]. For very small infants, a radiant warmer with a convective heat shield of thin polyethylene film provides a safe and effective thermal environment. An alternative approach would be the use of an incubator with added humidity to reduce evaporation, provided the infant's condition is stable enough that access for frequent procedures will not be required. Whenever possible, procedures such as blood sampling, starting intravenous infusions, and nursing care should be performed through the small ports in the incubator walls.

The major disadvantage of the radiant warmer is the increase in insensible water loss. With awareness of the problem and careful monitoring of water balance, the danger of dehydration can be minimized. Most evidence suggests that the metabolic heat production required of infants under radiant warmers is not appreciably higher than in incubators. The partition of heat loss is changed, but apart from the increased evaporative water loss this has no important implications for the infant's welfare.

Both incubators and radiant warmers have roles in the modern nursery. Radiant warmers are preferred for newly born, critically-ill and very small infants. Others can be easily managed in incubators.

Acknowledgements

Some of the studies described herein were supported by Basil O'Connor Starter Research Grant No. 5-268 and Clinical Research Grant No. 6-356 from the March of Dimes Birth Defects Foundation, and by grants-in-aid from Air-Shields, Inc. and Ohio Medical Products. Dr. Bell is also the recipient of New Investigator Research Award HD 16974 from the U.S. Public Health Service.

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